Security Fundamentals 8G4Z0015

Assessment ID: 1CWK100

Lecturer: Katie Paxton-Fear

Name: Summar Arshad

Student ID: 24820372

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Task 1: LinkedIn Blog

As a CEO of your company, you should be taking responsibility for the safety of your customers and staff. There are many cybersecurity attacks that occur daily to businesses that are of any size, no matter how large or small your company is. This can cause your business to deal with reputational and financial loss. A way to fix it would be to install an antivirus software that could mitigate most issues that you would usually come across in a computing system. This blog goes over the implementation of an antivirus software, why it is important, and consideration of the technical, legal, ethical, and organisational issues.

Why an Antivirus Software?

An antivirus software is important for the security system for a business as it is designed to detect and remove any malicious activity on the computer. This reduces the risk for any cyberattacks, which may include data breaches, ransomware, and spyware. For businesses that hold important information, such as medical records for a hospital, it is a legal requirement to implement an antivirus software into the IT system under the General Data Protection Regulation (GDPR). This law is a legal framework that sets out guidelines for the collection, processing, and the storage of any personal information from people who live in Europe. If a business that legally requires it to follow the GDPR law does not follow it, the fine framework can be around £17.5 million or 4% of their total global turnover annually. The EU GDPR sets a maximum fine of €20 million.

Breaches Reduced

Unethical hackers (hackers who exploit vulnerabilities illegally for personal gain or to disrupt operations) target businesses to get access sensitive data. When an antivirus software is supported by vulnerability scanning and an IT specialist, it helps mitigate these risks by identifying threats before they can cause any damage to the network. This helps data assets and information stay safe. An antivirus software can reduce the risk of many cyber-attacks, such as:

- **Malware Infections -** prevents malicious programs such as viruses, worms, and trojans from destroying any IT systems.
- **Ransomware Attacks -** detects and blocks any ransomware that could encrypt sensitive data.

- **Phishing Attacks -** identifies malicious email attachments and links designed to steal credentials or deploy malware.
- **Spyware -** stops software that covertly gathers sensitive information, such as login details and financial data.
- **Zero-Day Exploits -** offers protection against newly discovered vulnerabilities until patches are deployed.
- **Unauthorised Access -** prevents hackers from using any backdoor programs (such as PoisonTap) to gain access to the network or important information.

Why Should Businesses Use an Antivirus Software?

Most huge businesses, such as hospitals, legal firms, and financial organisations are legally required to use an antivirus software to protect any personal information. To the businesses that are not legally required to install one into the IT system, it is still recommended to many businesses. Many businesses avoid GDPR-related fines by implementing cybersecurity measures into their IT systems, and most of these include an antivirus software.

Implementing an antivirus software is the best cybersecurity tip for maintaining data integrity, safeguarding, and complying with the law. It shows that you care about your business by adding a layer of security to protect anyone involved within the business, such as customers, partners, and staff members. This upholds the reputation of most businesses as it shows that you care about the privacy and safety of your business.

There are many industries that by law, are required to install an antivirus software into their IT systems for cybersecurity measures. These include the healthcare industry, as HIPAA requires them to protect sensitive health information and patient privacy rights. There is also the PCIDSS, which is involved with the financial industry, where it also requires them to protect any personal data and report any breaches immediately. It keeps credit card data and transactions safe. As mentioned previously in this blog, GDPR also legally requires many businesses to safeguard any personal information and report breaches promptly.

It is recommended by the UK National Cyber Security Centre to conduct regular vulnerability scans and feedback for developers to address any issues that have been identified during those scans or the feedback. This makes it helpful for when a cybersecurity issue

needs to be fixed. It can be easily done with an antivirus software as it is able to scan any important information and report the information to a developer immediately if there are any issues that businesses can come across.

Alternatives

There are multiple alternatives that you can use instead of using an antivirus software.

1. **Online Virus Scan –** this scans your computer for any viruses or malware, just like an antivirus software but it is performed on the web, not installed on the PC. This is less secure than an antivirus and will not be able to scan everything on your PC.
2. **Firewall –** this is another secure alternative. This allows you to implement a firewall on your business' network. Once done, it will filter any malicious activity that may occur on your network. However, firewalls will not scan and remove any viruses and malware, even though it can protect your network from any malicious activity[1].
3. **Registry Cleaner –** this cleans up the registry of the Windows OS, it removes any redundant, invalid, or 'bad' entries. This is helpful for malware that target the Windows registry.
4. **Patch Management Software –** this is designed to help people manage their software updates. Rather than manually checking each application on your PC, it will automatically check it for you when it launches and allow you to download and install any updates when available after checking through your PC.
5. **Windows Defender –** this is the final alternative I am going to go over. This was originally released as an anti-spyware tool; however, it has been upgraded to remove any malware from the PC. It is included within the newest versions of Windows, so you will not need to install anything to turn it on. You can use this to scan for malware like an antivirus software, and it will remove it immediately.

How to Implement

Implementing the antivirus software into the IT system is very simple. You will firstly have to install a legitimate, paid antivirus software that is licensed. Paid antivirus software includes more features that would be suitable for a business. Most antivirus software

subscriptions are not expensive; however, it is still cost that the business should look into investing. Most antivirus software programs also update regularly to respond immediately to any new bugs.

The next step to implement it should be adding a new generation advanced endpoint security. This gives an immediate response that continuously checks if the person trying to enter is an attacker.

The final step would be to implement a DNS Web Content Filter, also known as web filtering. This adds a protective layer of defence before it reaches your business' internet. This prevents any exposure to malware and suspicious websites before any staff members try to access them.

Summary

An antivirus software is important for any IT system to keep information about the customers, business, and any stakeholders safe. With it being implemented, businesses can reduce the risks of any cyber-attacks. It also complies with any legal requirements if it is in the medical field, or a field where information must be kept confidential. It is a cheap investment that businesses, no matter how large or small, should look into implementing, and it has an overall long-term benefit that outweigh any negative reasons to not do it.

Task 2: Tutorial Blog

This is a tutorial that shows you how to use multiple operating systems on one PC. I will show you how to do this by setting up a Linux and Metasploit machine using an application called 'VirtualBox'. Linux is a free, open-source operating system that allows applications to run and is known for being secure. It is mainly used by programmers, as it is easier to setup a developer environment and keep up to date with the package management. Metasploit helps find security issues and verify vulnerability mitigations and is known to be one of the best penetration testing software that is currently available. With the 'VirtualBox' application, I will show you how to manage these systems on one device. Near the end of this tutorial, you should learn how to develop and use another OS for cybersecurity on a singular PC.

For this tutorial, you will need:

- **X86** hardware (Any Intel or AMD processor)
- At least **512 MB of RAM**
- Supported **host OS** (Windows, macOS, and Solaris)
- At least **30 MB** of hard disk space
- Supported **guest OS** (In this case, Linux)
- **VirtualBox** installation

Legal Considerations

- Make sure all OS and applications on VirtualBox belong to the application itself.
- If using the extension pack, make sure you have a commercial license.
- Make sure GDPR is being applied properly, or it will result in a fine.

Ethical Considerations

- Must be used for legal reasons and not unethical reasons.

Technical Considerations

- Network configuration, make sure not to use abridged mode or it will allow access for hackers to get into your network.
- Update the software regularly, just like any other application.
- Make regular backups for any situations where there may be a data loss.

Tutorial

Step 1:

To begin this tutorial, you will need to install VirtualBox using this link: https://www.virtualbox.org/.

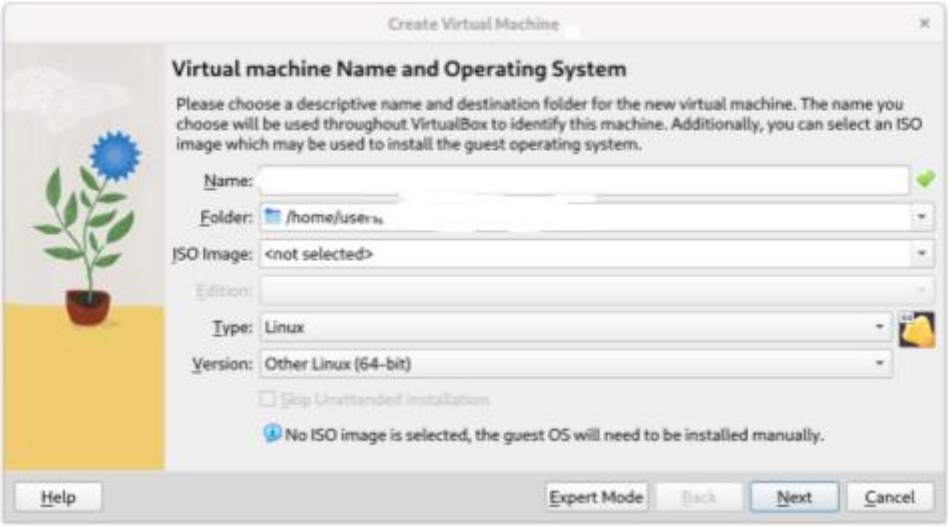
Step 2:

After installing VirtualBox, you will need to press the 'New' button as shown in 'Figure 1'. This will begin the installation of the Linux machine.

(Figure 1 - 'New' button on the Oracle VM VirtualBox Manager)

Step 3:

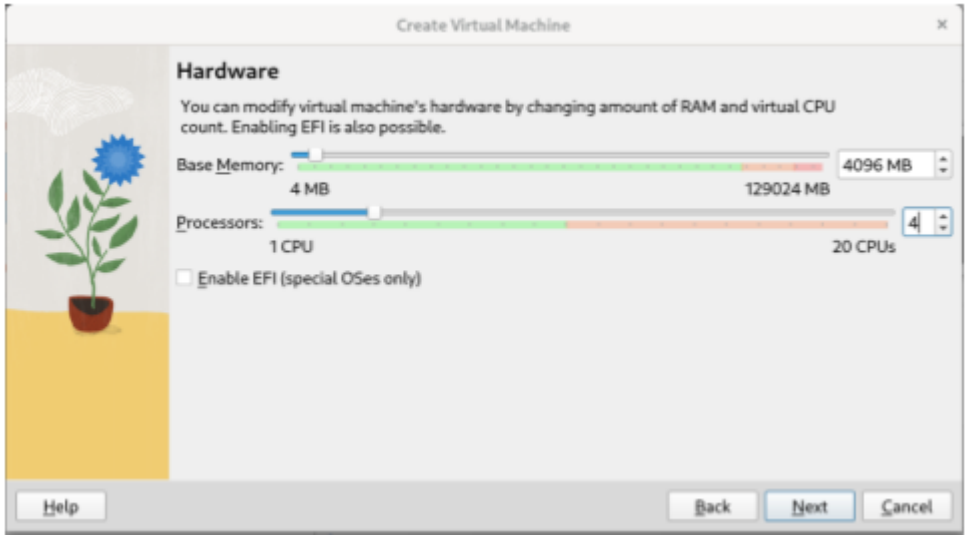
Name it anything you would like, but make sure you save it in the home directory so you can access it from other machines. This also makes sure that it is not saved in the hard drive. You should use the same exact settings that I have used down below.



(Figure 2 – Create Virtual Machine – Virtual machine Name and Operating System)

Step 4:

This is the step that manages hardware, you should change the ‘Processors’ section to ‘4’. You should also leave the tick box under ‘Enable EFI’ unticked.



(Figure 3 – Create Virtual Machine – Hardware)

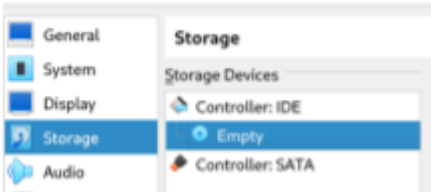
Step 5:

Press 'Do Not Add a Virtual Hard Disk'. You should then press continue after this step as we will not be installing one with a hard disk.

(Figure 4 – Create Virtual Machine – Virtual Hard disk)

Step 6:

Right click onto the machine you have just created and select 'Settings', then go on to the 'Storage' window.



(Figure 5 – Storage window)

Step 7:

Select a disk drive as we did not add one previously since you should be using a certain type of disk file for this installation. You should select the disk icon next to the optical drive. You should then press 'Other Locations', then go into 'VAR' then 'TMP'.

(Figure 6 – Optical disk file)

Step 8:

You should select the 'kali-linux-2021.1-live-amd64 iso' file that you have just installed onto the PC. This will be the hard disk drive.

(Figure 7 – 'kali-linux-2021.1-live-amd64 iso' file)

Step 9:

You should then press the option 'Enter', and that will mean that the kali machine has started up.

Step 10:

Go back to VirtualBox, and press 'New' to begin the installation of the Metasploit machine, as seen in 'Figure 1'.

Step 11:

Name the machine whatever you would like. Save it in the home directory so you can access it from other OS. You should also copy the settings I have done in ‘Figure 2’.

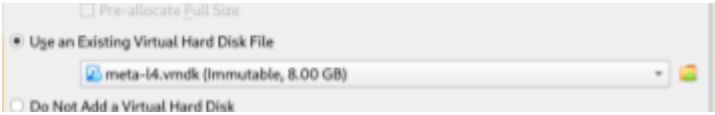
Step 12:

Change the ‘Base Memory’ manually and enter ‘2048 MB’, whilst keeping the ‘Processors’ as ‘1’.

(Figure 8 – Create Virtual Machine – Metasploit Hardware)

Step 13:

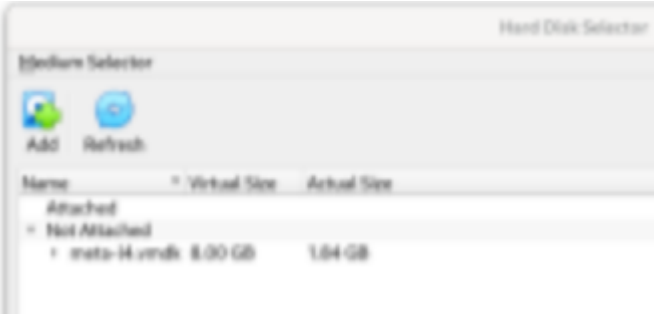
Select ‘Use an Existing Virtual Hard Disk’, but change the one that is already selected by pressing the folder image.



(Figure 9 – New file)

Step 14:

Press ‘New’ and go into the ‘Other Locations’ section which is shown in figure 6.



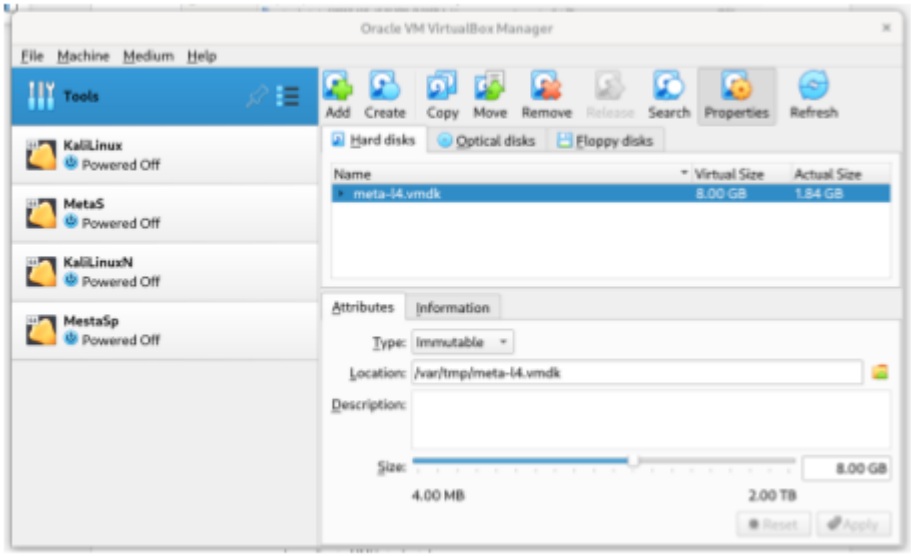
(Figure 10 – Hard Disk Selector)

Step 15:

Once clicked on ‘Other Locations’, click ‘/var’ then ‘/tmp’. This time, select the ‘meta-l4.vmdk’ file. Once you’ve done so, press 'File’ at the top left, 'Tools’, then ‘Media’.

Step 16:

You must make sure that it is set to ‘Immutable’ by pressing apply if it doesn’t already have it. After that, you can press ‘Start’ and you will have installed the Metasploit machine.



(Figure 11 – Oracle VM VirtualBox Manager – Tools)

This is the end of the tutorial. I hope you were able to successfully install VirtualBox, Linux, and Metasploit into a singular PC with no issues. If issues are persistent, you should head to https://www.virtualbox.org/manual/ for support.

References

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Task 3: Self-Reflection

TASK	POOR 0-39%	ACCEPTABLE 40-49%	GOOD 50-59%	VERY GOOD 60-69%	EXCELLENT 70-84%	OUTSTANDING 85%+
REPORT STRUCTURE AND REFERENCES (10%)	Lacking logical structure. Missing terminology from the field, or incorrect application of terminology. Blog posts are solely written by AI without synthesis from the student. References are AI hallucinations	Lacking structure, each sentence appears fragmented and unrelated points follow each other awkwardly. Language/style not appropriate for the audience (too technical, too informal…). An attempt at references but these are just URLs pasted into the document	Structured well, with terminology from the field applied correctly. Language/style appropriate for the stated audience, but may be suitable for a role which uses both technical and non-technical elements. An attempt at references	Use of headings and subheadings, all points present in some form. Language style at the appropriate level for the stated audience Some references.	All points present with correct formatting, an attempt to follow Harvard or IEEE references. Language style appropriate for the stated audience in a professional blog. Some academic references.	Correct formatting, Harvard or IEEE references and each figure is labelled correctly Language style appropriate for an executive audience in a professional blog and can be published without any stylistic changes. Proper academic sources and RFC's used.
SECURITY PRECAUTION (20%)	The security precaution discussed is not a recognised precaution and the article does not provide supporting evidence of this precaution's use in industry	Description copied and pasted from another source or uses many quotes to describe the precaution. Chooses a topic from the list of available topics.	An original description of the security precaution, with proper reference. Good why of a single security precaution, using a single source. Chooses a topic from the list of available topics.	A good summary of the security precaution directed at the appropriate technical level. 'Why' of a single security precaution, using multiple sources. Chooses a topic from the list of available topics, however, the topic has been expanded in some way.	A very good summary of the security precaution providing an executive with a comprehensive understanding of the security precaution. Evaluates alternative precautions in an informative way. Topic is original and not included on the list or has been significantly expanded or reworked.	Excellent summary of the security precaution for the audience not losing technical information Critical discussion between alternative precautions is presented, with the pros and cons of each listed. Topic is original and not included on the list of examples.

Report Structure and References (10%)

- Use of headings and subheadings
- Appropriate language for the audience
- Harvard references

Security Precaution (20%)

- Good summary of the security precaution
- Using appropriate language for the audience
- Evaluates alternative precautions in an informative way
- Topic is chosen from the list of available topics, however it has been expanded

LEGAL AND ETHICAL ISSUES (10%)	No discussion of the ethical or legal considerations involved in the security precaution No discussion of issues within organisations.	Some discussion of legal issues, however this is limited in scope or copied and pasted. No explicit reference to the practicality of policy or procedural concerns of implementing the security precaution.	Clear discussion of legal and ethical issues but without an explicit reference to a law or piece of ethics. An original description of the legal and ethical issues. Some reference to how this could be implemented in terms of policy or procedure	Clear reference to a piece of relevant UK or US legislation such as the Data Protection Act, the Computer Misuse Act, Regulation of Investigatory Powers Act or Computer Fraud and Abuse Act. Discusses the practicality of implementing the security precaution	Reference to a relevant piece of ethical considerations such as a textbook on ethics or an article written about an ethical concern in cyber security. Discusses the practicality including any organisational requirements such as a large vs small org.	A discussion of the conflicting requirements of legal concerns and ethical concerns in cyber security. Analysis of the cost and benefit of implementing this within an organisation including methods for reducing the organisation cost
FUTURE ATTACK RISK (10%)	No information on why or how this precaution can reduce the risk of a future attack.	Some information on how this security control reduces the risk of attack, but limited discussion of how.	Clear guidance on why and how this particular security control reduces the risk of an attack.	Clear guidance on why and the specific types of attack with security control prevents. Uses an example, but this is a theoretical attack.	Clear 'how' and 'why' this security precaution can prevent a specific type or group of attacks. Uses an example attack from the news, this may be a historical attack.	Clear 'how' and 'why' this security precaution can prevent attacks, including a discussion on what type of attacks it cannot reduce the risk of. Uses a recent (within the previous 2 years) example attack from the news.
TOOL CHOICE & CODE (10%)	No tool is discussed or presented. Screenshots presented are copied from the lab worksheets, rather than your own work.	Some tooling is discussed, but it is not clear which exact tool.	Tool was covered in class or is one of the examples from the assessment.	Tool was covered in class or is one of the examples.	Tool was covered in class or is one of the examples, with some mention of workflow between tools, e.g. Nmap to Metasploit	Tool was not covered in class.

Legal and Ethical Issues (10%)

- Reference to a relevant piece of ethical consideration
- I have used an article written about my ethical concern in this assignment
- Discusses practicality of implementing the security precaution
- Compare large vs small organisation?

Future Attack Risk (10%)

- Clear ‘how’ and ‘why’ this security precaution can prevent attacks
- Used an example from the media

Tool Choice & Code (10%)

- Tool was covered in class
- One of the examples

TUTORIAL (20%)	Tutorial missing.	Limited tutorial, demonstrating a non-primary feature. Some link to further reading.	Very good tutorial which demonstrates the tools primary feature Clear link to further reading, with some scaffolding "to learn more about x features you can read y article" Limited case for 'why' this tool.	Clear tutorial which presents the core feature of tool with some discussion of alternative features. Link to further reading for alternative features. A strong 'why' case is presented for the tool	Clear tutorial presenting multiple features of the tool. Link to further reading for each feature linking to another tutorial, book, or documentation Tutorial discusses alternative tools to achieve the same result	Very in-depth tutorial, demonstrating multiple features of the tool in great technical detail. References to further reading to specific documentation providing Tutorial critically analyses alternative tools.
SCREENSHOTS (10%)	No screenshots in the tutorial.	Screenshots may be too small to read but are present, or the screenshots are cut making them difficult to read	Screenshots can be read easily and at a reasonable zoom level.	Screenshots have been cropped and are easily read at a reasonable zoom level, with relevant information highlighted and/or visible.	Screenshots are clear, highlight only the important information and have appropriate labels under each.	Screenshots are labelled and are referred to in the text with 'Figure 1', 'Figure 2'
SELF REFLECTION (10%)	No annotated copy of the marking grid	An annotated copy of the marking grid but with some annotations missing.	Annotated copy of the marking grid without any supporting comments.	A copy of the marking grid with some supporting comments Comments explaining how a student used generative AI in addition to their own work	A copy of the marking grid with a bullet point for each annotation, easy to read and organised. Evidence of an honest self-reflection.	Additional self reflection on the unit and what was learned outside of the assessment only, this can be critical of the unit.

Tutorial (20%)

- A clear tutorial that presents the core features of the tool
- Included alternative features

- Further information link provided
- Why is it necessary to use the tool

Screenshots (10%)

- Screenshots include clear, necessary information
- Screenshots include labels

Self-reflection (10%)

- A copy of the marking grid
- Bullet points for each section
- Simple and organised

I think I have covered most important points that were to be included in this assignment. I had missed a few areas, and I had not done a few things that were required in the specification.